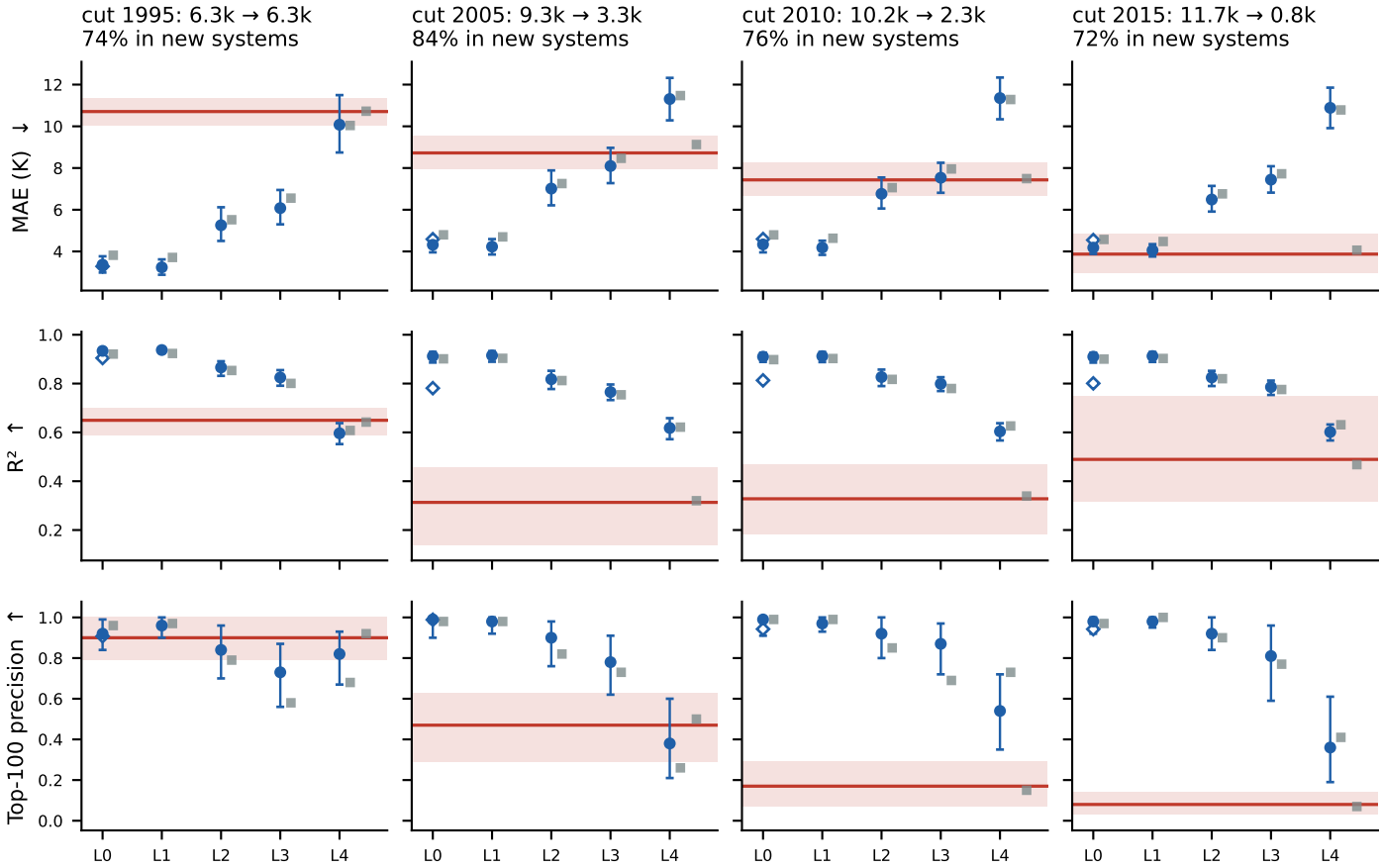
Random validation (L0/L1) overstates R^2 on superconductors reported after the cut at every cut; grouped protocols (L2–L4) come closer but none anticipates it stably



realised performance on the temporal split (XGBoost); band = 95% family-bootstrap interval

retrospective out-of-fold estimate of each protocol (XGBoost)

L0 re-weighted to the new/seen-system mix of the later compositions

RandomForest (protocols; realised value at the right)